Predicting the likelihood of Type 2 Diabetes

Team 3c

2026.02.01

CDS-492 | Spring 2026 | Dr. Slamani

Module 01 Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza

Outline

- MD01 – First Team Meeting
  - Define roles
    - Team Lead: Tyson Johnson
    - Note taker/task scheduler: Nithila Neethi Devan
    - Assignment reviewer/submitter : Alizeh Murtaza
  - Discuss choices and ranking of the two project topics
    - Topic J : How can we use patient data to predict the likelihood of developing Type 2 Diabetes?
    - Topic H: How can a hospital optimize the allocation of its resources to improve patient outcomes?
  - Submit assignment

Meeting Minutes

Date: 01/24/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza

Agenda:

- Discuss positions/roles within the team
- Discuss data source options
- Future Assignment Planning

Notes:

- Team Selected Topic J (Type 2 Diabetes) as project focus
- Potential Data Source: South Korean Government Data Portal
  - Tyson has existing logic to translate data via Python to English.
- Literature Review Requirement: Each member should find 2-3 sources prior to next meeting; we can discuss strongest sources/what to include during the next meeting.
- Team has decided to continue with virtual meetings; next meeting is set for Jan 31st at 5pm (flexible)

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 2-3 sources for the literature review for discussion during next meeting.	Tyson, Nithila, Alizeh	01/24/2026	Started
T02					
T03					
T04					
T05					
T06					
T07					

Note:

- Status: Not started, Started, Waiting on Name of person, Date Completed
- For more tasks, add rows to the table by right clicking on a cell in the last row and choose from the horizontal menu “Insert Rows Below”
- Keep the same table for the following week(s) and add more rows

Module 02

Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza

MD02 – Revisions to work from previous module(s)

- Updated title from given question to: Predicting the likelihood of Type 2 Diabetes (Slide 1)
- Updated Task Description and Status (Slide 9)
- Minor Formatting Changed (Slide 4, Slide 5, Slide 8, Slide 9)

Meeting Minutes

Date: 02/01/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza

Agenda:

- Review first submission
- Discuss slide formatting
- Discuss literature review articles
- Complete "Data Search" slide
- Update tasks
- Schedule next meeting

Notes:

- Discussed Potential Data Sources and Factors to consider when finding data.
- Ask Professor if our question can be slightly modified to accommodate non-time-series data if longitudinal data are not publicly available.

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 1 reliable source and complete our literature review and synthesis matrix on.	Tyson, Nithila, Alizeh	01/24/2026	Completed
T02	Research Data Options	Find Data that tracks a unique identifier information, data that would allow for time series methods to be used.	Tyson, Nithila, Alizeh	01/31/2026	Started
T03					
T04					
T05					
T06					
T07					

Note:

- Status: Not started, Started, Waiting on Name of person, Date Completed
- For more tasks, add rows to the table by right clicking on a cell in the last row and choose from the horizontal menu “Insert Rows Below”
- Keep the same table for the following modules(s) and add more rows

MD02 – Literature Review | Paper 1

- Paper title: The diabetes risk score: A practical tool to predict type 2 diabetes risk
- Paper citation:
  - Lindstrom, J., & Tuomilehto, J. (2003). The Diabetes Risk Score: A practical tool to predict type 2 diabetes risk. *Diabetes Care*, *26*(3), 725–731. https://doi.org/10.2337/diacare.26.3.725
- Summary of what was investigated:
  - Study investigated whether type 2 diabetes risk can be predicted using noninvasive patient data without laboratory tests
- Results:
  - Identified key predictive variables including age, BMI, waist circumference, history of hypertension treatment, history of elevated blood glucose, physical activity, and diet.
  - The resulting Diabetes Risk Score demonstrated strong predictive performance (AUC ≈ 0.85–0.87), with scores of 9 or higher indicating substantially increased risk.
- What processing algorithms did each paper use:
  - Multivariate logistic regression, Regression coefficients converted to categorical point-based risk scoring, ROC curve analysis
- What you liked about the paper:
  - Model relies on accessible, real-world patient data, making it practical for large-scale screening.
- What additions you can provide to the topic:
  - Include additional patient data such as family history, socioeconomic factors, and race/ethnicity

MD02 – Literature Review | Paper 2

- Paper title:
  - AUSDRISK: an Australian Type 2 Diabetes Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures
- Paper citation:
  - Chen, L., Magliano, D. J., Balkau, B., Colagiuri, S., Zimmet, P. Z., Tonkin, A. M., Mitchell, P., Phillips, P. J., & Shaw, J. E. (2010). AUSDRISK: an Australian Type 2 Diabetes Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures. *Medical Journal of Australia*, *192*(4), 197–202. https://doi.org/10.5694/j.1326-5377.2010.tb03478.x
- Summary of what was investigated:
  - The development and design of AUSDRISK.
- Results:
  - AROC: 0.78 (95% CI, 0.76–0.81), HL Chi-Squared: 4.1 (P = 0.85)
  - Sensitivity: 0.74, Specificity: 0.677, PPV: 0.127
- What processing algorithms did each paper use:
  - Multiple linear regression, scoring system relying on beta scores. AROC and chi-squared for validation.
- What you liked about the paper:
  - Explained a lot of their decision-making process on things like certain features' inclusion/exclusion.
- What additions you can provide to the topic:
  - By using a different dataset and similar methodology, we can strive to reproduce similar results.

MD02 – Literature Review | Paper 3

- Paper title:
  - Opportunistic screening of type 2 diabetes with deep metric learning using electronic health records
- Paper citation:
  - Jin, Q., Zhang, H., Szczerbinski, L., Zhu, J., Gerych, W., Xu, X., Wang, K., Hsu, S., Mandla, R., Deutsch, A. J., Manning, A., Mercader, J. M., Hartvigsen, T., Udler, M. S., & Ghassemi, M. (2025). Opportunistic screening of type 2 diabetes with deep metric learning using electronic health records. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-25759-x
- Summary of what was investigated:
  - Researchers studied whether existing patient health records could be used to predict who will develop type 2 diabetes in the future using ML models.
- Results:
  - The ML model predicted Tupe 2 diabetes more accurately than traditional statistical methods. It was able to identify high risk patients earlier by detecting the various patterns that can lead to patients developing diabetes.
- What processing algorithms did each paper use:
  - Deep Metric Learning trained on Electronic Health Records
- What you liked about the paper:
  - It combines both predicting diabetes onset and identifying patient subtypes in one model, this allows for more personalized pattern detecting, and therefore interventions
- What additions you can provide to the topic:
  - The incorporation of lifestyle or social factors to improve prediction accuracy.
  - Use of explainable AI to make predictions and pattern detection process more transparent and interpretable.

MD02 – Synthesis Matrix | Paper 1

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Lindström, J., & Tuomilehto, J. (2003). The diabetes risk score: A practical tool to predict type 2 diabetes risk. Diabetes Care, 26(3), 725–731.	The paper addresses whether type 2 diabetes risk can be accurately predicted using noninvasive, easily collected patient data. The authors hypothesize that a simple risk score, based on demographic, anthropometric, behavioral, and medical history variables, can identify individuals at high risk for developing drug-treated type 2 diabetes.	Population-based cohort study using Finnish FINRISK surveys (1987 and 1992). Participants: adults aged 35–64 without antidiabetic drug treatment at baseline. Follow-up: 5–10 years using national drug registry data. Multivariate logistic regression used to identify significant predictors.	A risk score ≥9 effectively identified individuals at high risk for future drug-treated type 2 diabetes. The model demonstrated strong predictive performance (AUC ≈ 0.85–0.87). The score also performed well in identifying undiagnosed diabetes in cross-sectional analyses.	Uses noninvasive, low-cost, easily accessible patient data. Strong validation across independent cohorts. Practical and scalable for population-level screening.	Study population limited to Finnish adults, reducing generalizability. Outcome defined as drug-treated diabetes, potentially missing earlier or diet-controlled cases. Use of categorical scoring may reduce predictive precision.

MD02 – Synthesis Matrix | Paper 2

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Chen, L., Magliano, D. J., Balkau, B., Colagiuri, S., Zimmet, P. Z., Tonkin, A. M., Mitchell, P., Phillips, P. J., & Shaw, J. E. (2010). AUSDRISK: an Australian Type 2 Diabetes Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures. *Medical Journal of Australia*, *192*(4), 197–202. https://doi.org/10.5694/j.1326-5377.2010.tb03478.x	"To develop and validate a diabetes risk assessment tool for Australia based on demographic, lifestyle and simple anthropometric measures." The authors explain their development of AUSDRISK.	9 risk factors categorized Pearson chi-squared test Prediction models made using multiple linear regression. Risk factors: age, sex, ethnicity, parental history of diabetes, history of high blood glucose level, use of antihypertensive medications, smoking, physical inactivity, waist circumference	AusDiab data: 1999-2000, 11,247 adults, Ages 25+, Australian 5-year follow-up: 2004-2005, 6,537/10,788 surviving returned Used 6,060 participants after removing 246 unclassifiable and 229 baseline diabetic (from AusDiab). AROC: 0.78 (95% CI, 0.76–0.81) HL Chi-Squared: 4.1 (P = 0.85) Sensitivity: 0.74 Specificity: 0.677 PPV: 0.127	The data used included a wide variety of features measured that other models may not have used, such as the use of certain medications and parental history of diabetes. The source provides a working example of a model used for predicting T2D. Can be accessed as a free "online interactive risk assessment tool." http://health.gov.au/internet/main/publishing.nsf/Content/diabetesriskassessmenttool	Acknowledges that data limitations would likely result in overly raised risks in ages under 25. May have also benefitted from a more ethnically diverse pool of participants for broad use.

MD02 – Synthesis Matrix | Paper 3

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Jin, Q., Zhang, H., Szczerbinski, L., Zhu, J., Gerych, W., Xu, X., Wang, K., Hsu, S., Mandla, R., Deutsch, A. J., Manning, A., Mercader, J. M., Hartvigsen, T., Udler, M. S., & Ghassemi, M. (2025). Opportunistic screening of type 2 diabetes with deep metric learning using electronic health records. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-25759-x	Can routine electronic health records be used to predict the future onset of type 2 diabetes and identify patient subtypes?	Trained a deep metric learning model on longitudinal EHR data to predict future Type 2 diabetes onset and learn patient patterns Used these learned patterns to cluster patients into distinct subtypes Compared model performance with logistic regression and other baseline methods	The DML model outperformed traditional prediction methods for future T2D onset prediction and revealed three distinct subtypes of patients with different clinical characteristics and responses to treatment. AUROC of *0.969* Sub types : Green (healthier, more responsive), Yellow (intermediate), Red (high comorbidity burden)	Combines disease prediction and patient subtyping within a single model Utilizes large, diverse EHR datasets Enables practical screening using existing clinical data without additional testing	Does not include lifestyle or social determinants of health Limited interpretability due to the black-box nature of deep learning

MD02 – Data Search

- Data sources
  - https://www.data.go.kr/en/data/15007122/fileData.do
- Relevance of the data to the project
  - Dataset contains features that is publicly available.
  - Dataset contains yearly health exam results of South Korean patients.
- Possible challenges with the data
  - Data is specific to the Korean demographic
  - Unsure if there is a way to incorporate timeseries methods as we are still identifying data with unique identifier information (subscriber serial number)